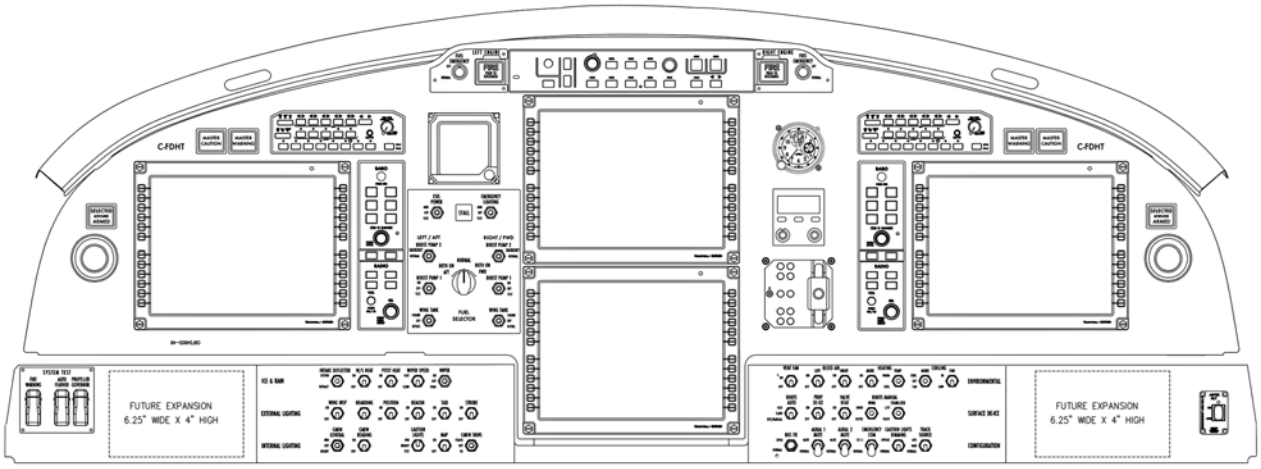
Display Units – General Data

1. General

The Apex Avionics suite is based on a four Display Unit (DU) layout arranged in a T configuration, refer to Figure 1.



Primus Apex Display Units (T–Configuration)
Figure 1

The displays are manufactured by Esterline/Barco for Honeywell. Replacement displays must be designated Honeywell KDU 1080 (Pre Mod 6/2450) or KDU 1080-2 (Post Mod 6/2450) display units. The aircraft must be configured with the same model of display units in all four locations. Intermixing of KDU-1080 and KDU-1080-2 display units is not an approved configuration.

Additional descriptive material and operating instructions are provided in the DHC-6 Series 400 Pilot Operating Handbook (POH), PSM 1-64-POH, Section 7 and in the Honeywell Apex Pilot's Guide.

2. Description

A. Display Units (Honeywell Model KDU-1080 or KDU-1080-2)

There are four interchangeable displays units installed on the instrument panel:

DU 1 – Pilot's Primary Flight Display, PFD 1

DU 2 – Upper Multi-Function Display, MFD 1

DU 3 – Lower Multi-Function Display, MFD 2

DU 4 – Co-pilot's Primary Flight Display, PFD 2

The DU is a 10 x 8 inch (10.4 inch diagonal display area) LCD display screen with integral LED backlighting. The display unit does not contain any system software. The display units are driven by the Apex AGMs installed in the MAU (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data). The display screen as currently installed on the aircraft does not contain integral heaters. Optional installations may be available in the future that approve this option.

The display screen is divided into 1/6 or 2/3 size sections referred to as windows. Each window has a default display and functionality defined. Each window functions independently from the other windows. This allows

multiple system operation and functions to be displayed on one DU at the same time, and also to maintain independence and a high degree of software reliability in each window.

Each type of display unit has a designated controller which allows the operator to navigate within the windows on the screen and set the format for the system information to be displayed. For the PFDs, the associated controllers are Primary Flight Display controllers, and the two MFDs are controlled through the Multi-Function Controller.

Bezel lighting on each DU is controlled by dimming potentiometers labelled INSTRUMENT in the centre pedestal controller mounted on the control column near the MFD Controller.

Display brightness is controlled by the Display Reversion Control Panel. The brightness of each DU may be controlled independently of the other DUs.

B. Primary Flight Displays (PFDs)

The Pilot's PFD is installed on the left side of the instrument panel. 28 Volt DC power is provided from the left bus through a 5 ampere PILOT PFD (H1) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The Co-pilot's PFD is installed on the right hand side of the instrument panel. 28 Volt DC power is provided from the right bus through a 5 ampere COPILOT PFD (H2) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

C. Multi-Function Display (MFD)

The upper MFD is installed in the center of the instrument panel. The Upper MFD default format is the Situational Awareness Display.

28 Volt DC power is provided from the left bus through a 5 ampere UPPER MFD (J1) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The lower MFD is installed below the upper MFD in the centre of the instrument panel. The Lower MFD default format is the Systems Display.

28 Volt DC power is provided from the right bus through a 5 ampere LOWER (MFD) (J2) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

3. Adjustment/Test

A. Power-up

1) Power On

a) Connect external power to the aircraft (refer to 12-00-00, Servicing – General Data).

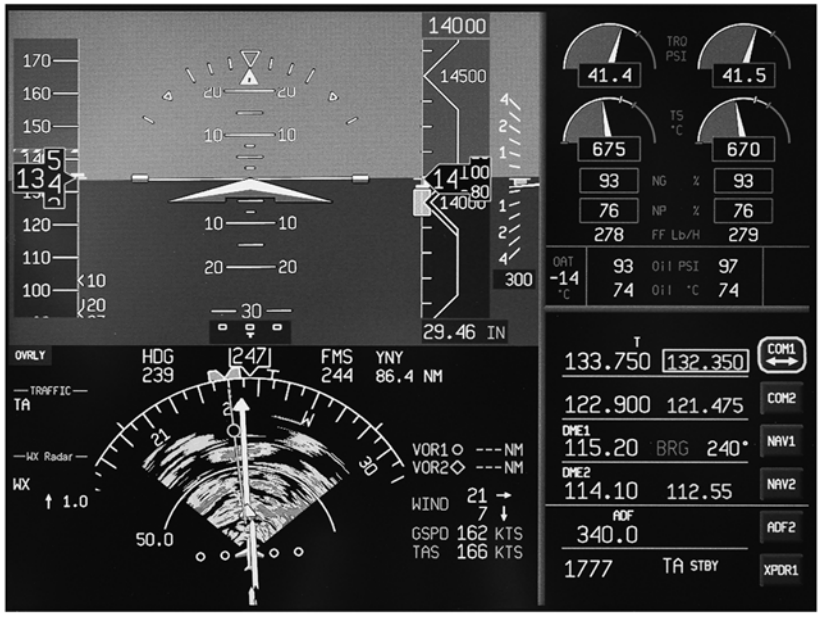
b) Ensure that the switches on the overhead panel are set as follows:

1 Select POWER SOURCE switch to EXTERNAL and MASTER switch to ON (Pre Mod 6/2314); or EXTERNAL switch to ON (Post Mod 6/2314).

c) Ensure that PFD 1, MFD 1, MFD 2 and PFD 2 come on.

Note Boot-up of the Apex system may take up to 24 seconds.

B. Left Primary Flight Display (PFD 1) (Pilot's)



Primary Flight Display
Figure 2

1) Ensure that the pilot's (left) PFD is on and shows:

   - ADI and HSI
   - Engine window
   - Radio window

2) Ensure that left PFD controller is working and can do its control functions on the PFD. Test functions such as navigation source select (including FMS), tuning VHF communication and navigation systems such as COM 1, COM 2, NAV 1, NAV 2, and tuning other systems such as ILS, ADF 1, ADF 2, DME 1, DME 2, Transponder 1, and Transponder 2. (Refer to Figure 2).

   Note Not all aircraft will have all of this equipment installed.

3) Using Reversion Control Panel, confirm pilot's PFD can be dimmed and that it can be turned off.

4) Using Reversion Control Panel, revert the pilot PFD to AGM 2.

5) Ensure the pilot PFD display matches the co-pilot PFD, and the co-pilot PFD reverts to the pilot PFD display format (engine and radio window on the RH side).

C. Upper Multi-Function Display (Upper MFD)

1) Ensure that the upper MFD is on and shows:

   a) Situational Awareness Display

2) Ensure that the MF controller is working and can focus on various displays on the upper MFD. Ensure bezel buttons, where enabled, are functioning.

3) Using Reversion Control Panel, confirm upper MFD can be dimmed and that it can be turned off.



Situational Awareness Display
Figure 3

D. Lower Multi-Function Display (Lower MFD)



Systems Display
Figure 4

1) Ensure that the lower MFD display is on and shows:

- The systems windows
- Fuel system window including left wing fuel quantity, aft fuselage belly tank fuel quantity, forward fuselage belly tank fuel quantity, and right wing fuel tank quantity (wing tanks are optional)
- Left and right generator voltages
- Left and right generator currents

- Hydraulic system pressure
- Brake system pressure
- Electronic checklist (ECL) (if option installed)
- CAS message window
- Ice protection system annunciations (if ice protection system installed)
- Cabin and cockpit temperatures
- Battery voltage and current

2) Ensure that the MF controller is working and can focus on various displays on the lower MFD.

3) Using Reversion Control Panel, confirm lower MFD can be dimmed and that it can be turned off.

E. Right Primary Flight Display (PFD 2) (Co-Pilot's)

Note The co-pilot's PFD is the mirror image, in terms of screen arrangement, from the pilot's (left) PFD (refer to Figure 2).

1) Ensure that the co-pilot's PFD is on and shows:

- ADI and HSI
- Engine window
- Radio window

2) Ensure that: right PFD controller is working and can do its control functions on the PFD. Test functions such as navigation source select (including FMS), tuning VHF communication and navigation systems such as COM 1, COM 2, NAV 1, NAV 2, and tuning other systems such as ILS, ADF 1, ADF 2, DME 1, DME 2, Transponder 1, and Transponder 2.

Note Not all aircraft will have all of this equipment installed.

3) Using Reversion Control Panel, confirm co-pilot's PFD can be dimmed and that it can be turned off.

4) Using Reversion Control Panel, revert the co-pilot PFD to AGM 1.

5) Ensure the co-pilot PFD display matches the pilot PFD (engine and radio window on the right side).

F. Miscellaneous Checks

1) Window Navigation and Focus

a) On the Multi-Function Controller (MF Controller), press the left arrowhead button.

b) Ensure that window focus (a cyan coloured box border) moves to the next available interactive window left of the current window having focus.

c) On the MF Controller, press the down arrowhead button.

d) Ensure that window focus moves to the next available interactive window below the current window having focus.

e) On the MF Controller, press the up arrowhead button.

f) Ensure that window focus moves to the next available interactive window above the current window having focus.

g) On the MF Controller, press the right arrowhead button.

h) Ensure that window focus moves to the next available interactive window to the right of the current window having focus.

i) Wait approximately thirty seconds. The window focus is removed after 30 seconds.

2) Cursor Display Check

a) On the MF Controller, press the COM button to move the window focus to the PFD 1 radio window.

b) Ensure that the cursor focus defaults to COM 1.

c) Use the MF Controller joystick and ensure that you can move the cursor focus to:

- COM 1
- COM 2

d) Put the cursor focus on XPDR and wait for approximately twenty seconds.

e) Ensure that the cursor focus moves to the first field that can be modified in the current window.

f) Use the PFD bezel buttons to select the different functions in the radio window. Ensure that you can select the different functions.

3) Detail Window Display

a) Ensure that the PFD radio window has window focus, and ensure that COM 1 (or COM 2) has cursor focus.

b) Press the DETAIL button on the PFD 1 Controller. Ensure that the COM details page opens. Press the ENT button on the MF Controller, or the SEL knob on the PFD Controller and ensure that the entered value is selected/activated.

c) Press the DETAIL button again and ensure that the:

- Details page closes
- Radio window opens
- Cursor focus is on the radio selected on the details page

G. Close Up

1) On the overhead panel, switch OFF:

a) Select MASTER and POWER SOURCE switch to OFF (Pre Mod 6/2314); or EXTERNAL switch to OFF (Post Mod 6/2314).

2) Disconnect external power source from aircraft (refer to 12-00-00, Servicing – General Data).

4. Removal and Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – PANEL/PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

Note The Display Units have additional chassis grounding requirements, refer to the Note in Para 4.A.3).

A. Primary Flight Display (PFD)

1) General

These procedures apply to the pilot and co-pilot's PFD.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open (5 ampere PILOT PFD (H1) for the left PFD or 5 ampere COPILOT PFD (H2) circuit breaker for the right PFD) circuit breaker and fit with safety clip.

c) Remove the screws that hold the PFD in the instrument panel.

d) Carefully pull the PFD out of the instrument panel to get access to the electrical connectors (refer to PSM 1-64-2W) on the PFD.

e) Disconnect the electrical connectors from the PFD.

f) Remove the PFD from the aircraft.

Note Stow connectors as required.

3) Installation

Note Instrument panel surface area under the display unit is used for electrical bonding of the display unit. Ensure bonding surface between display and instrument panel is clean and not contaminated. Clean if required to ensure bonding.

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the (5 ampere PILOT PFD (H1) for the left PFD or 5 ampere COPILOT PFD (H2) circuit breaker for the right PFD) circuit breaker is opened and fitted with a safety clip and the connectors are stowed as required.

c) Hold the PFD (refer to Figure 1 and Figure 2) near the instrument panel and connect the electrical connectors.

d) Put the PFD into the instrument panel.

e) Insert the screws that hold the PFD in the instrument panel. If screws are damaged, replace.

f) Tighten the screws.

g) Remove safety clips and close (5 ampere PILOT PFD (H1) for the left PFD or 5 ampere COPILOT PFD (H2) circuit breaker for the right PFD) circuit breaker on the avionics circuit breaker panel.

h) Complete the applicable adjustment/test procedure, refer to Para 3.

B. Multi-Function Display (MFD)

1) General

These procedures apply to the upper and lower MFD.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) On the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data), open and safety the affected circuit breakers (5 ampere UPPER MFD (J1) for the upper MFD or 5 ampere LOWER MFD (J2) circuit breaker for the lower MFD).

c) Remove the screws (refer to Figure 1 and Figure 3 for the upper MFD or Figure 4 for the lower MFD) that hold the MFD in the instrument panel.

d) Carefully pull the MFD out of the instrument panel to get access to the electrical connectors (refer to PSM 1-64-2W) on the MFD.

e) Disconnect the electrical connectors from the MFD.

Note Stow connectors as required.

f) Remove the MFD from the aircraft.

Note Ensure that the MFD screen and connectors are protected during the removal process.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the (5 ampere UPPER MFD (J1) for the upper MFD or 5 ampere LOWER MFD (J2) circuit breaker for the lower MFD) circuit breaker is opened and fitted with a safety clip and the connectors are stowed as required.

c) Hold the MFD (refer to Figure 1 and Figure 3 for the upper MFD or Figure 4 for the lower MFD) near the instrument panel and connect the electrical connectors.

d) Put the MFD into the instrument panel.

e) Insert the screws (refer to Figure 3 for the upper MFD or Figure 4 for the lower MFD) that hold the MFD in the instrument panel. If screws are damaged, replace.

f) Tighten the screws.

g) Remove safety clips and close (5 ampere UPPER MFD (J1) for the upper MFD or 5 ampere LOWER MFD (J2) circuit breaker for the lower MFD) circuit breaker on the avionics circuit breaker panel.

h) Complete the applicable adjustment/test procedure, refer to Para 3.

5. Troubleshooting

A. Primary Flight Display (PFD)

Note A fault with the Display Reversion Control Panel may cause the symptoms of a faulty display. Refer to 31-15-00, Display Reversion Control Panel – General data, Para 5. for Display Reversion Control Panel troubleshooting.

1) Ensure PILOT PFD (H1) & COPILOT PFD (H2) circuit breakers are closed.

2) If presented with any of the following CAS messages:

- Check DU 1 (Caution)
- Check DU 4 (Caution)
- Check Pilot PFD (Caution)
- Check Copilot PFD (Caution)

a) Use the associated DU brightness knob on the reversion panel to turn the affected display completely off (past the final detent).

b) Wait several seconds, then turn the display back on.

c) If CAS message persists, the associated PFD may require service (refer to Para 5.A.7)).

3) If display is blank:

a) Ensure associated DU brightness knob is not in the OFF position.

b) Do an AGM reversion for the affected display.

c) If display remains blank, it may require service (refer toPara 5.A.7)).

d) If display mirrors the opposite PFD, the issue may be in the wiring or the associated AGM.

4) Visually inspect the electrical connectors for damaged wiring or contacts. Ensure connectors are fully seated and tight.

5) Swap PFD with a known good unit. If issue clears, the removed PFD requires service (refer to Para 5.A.7)).

6) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

7) If repair or service is required, remove and send to an appropriately rated avionics repair facility.

B. Multi-Function Display (MFD)

Note A fault with the Display Reversion Control Panel may cause the symptoms of a faulty display. Refer to 31-15-00, Display Reversion Control Panel – General data, Para 5. for Display Reversion Control Panel troubleshooting.

1) Ensure UPPER MFD (J1) & LOWER MFD (J2) circuit breakers are closed.

2) If presented with any of the following CAS messages:

- Check DU 2 (Caution)

- Check DU 3 (Caution)

a) Use the associated DU brightness knob on the reversion panel to turn the affected display completely off (past the final detent).

b) Wait several seconds, then turn the display back on.

c) If CAS message persists, the associated MFD may require service (refer to Para 5.B.7)).

3) If display is blank:

a) Ensure associated DU brightness knob is not in the OFF position.

b) Do an AGM reversion for the affected display (DU1 outer knob for DU 2, DU 4 outer knob for DU 3).

c) If display remains blank, it may require service (refer to Para 5.B.7)).

d) If display shows a red X, the issue may be in the wiring or the associated AGM.

4) Visually inspect the electrical connectors for damaged wiring or contacts. Ensure connectors are fully seated and tight.

5) Swap MFD with a known good unit. If issue clears, the removed MFD requires service (refer to Para 5.B.7)).

6) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

7) If repair or service is required, remove and send to an appropriately rated avionics repair facility.